

# Public Economics and Development

## Week 1: Course Introduction

Seung-hun Lee

Taipei School of Economics  
National Tsing Hua University

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# Course logistics

Instructor: Seung-hun Lee

- Office location: Room C06 in this building
- Office hours: Tuesdays 2:00-4:00PM ([Sign up here](#))
- Ph.D. in Economics from Columbia University (2024)

Class time and location

- Time: Tuesdays 9AM - 12PM
- Location: Room A09

Coming prepared to the lecture

- Lecture notes are uploaded a day in advance
- Slides summarize textbooks and papers – no need to purchase them separately
- Keep up with the readings: more on that later

# Course goal: Understand public economics in developing countries

Study government's role in economic life: Taxation, public goods, and bureaucracy

- Classical theories on optimal taxation, redistribution, and public sector employment
- Income taxes, antipoverty programs, principal-agent problems etc.

Understand challenges faced by developing countries

- Lagging behind on fiscal, legal, and personnel capacity of the state
- Thus, what's optimal elsewhere do not work on developing countries

Goals and requirements

- Be able to critically engage with scholarly work
- Conduct independent, original research
- Familiarity with statistical inference, micro/macroeconomics is necessary!

# Course goal: Understand public economics in developing countries

## Main texts

- Atkinson, Stiglitz (2015): Lectures on Public Economics
- Besley, Persson (2009): Pillars of Prosperity
- Hindriks, Myles (2013): Intermediate Public Economics

## Recommended (and often helpful texts)

- Public economics: Gruber (2022, undergrad), Roland (2014, undergrad), Salanié (2011, graduate)
- Econometrics: Angrist, Pischke (2009, graduate), Cameron, Trivedi (2005, graduate), Stock, Watson (2020, undergraduate)

## Readings

- 4-6 papers from leading economic scholarly journals
- Read up on the starred papers in the reading list!

# Course topics (Refer to syllabus for full reading list)

## Weeks 1-2: Introduction and econometric methods

- Introduction to public economics
- Empirical analysis tools: OLS, IV, Panel regression, Natural experiments

## Week 3-5: Classical theories on taxation and public economics

- How drastic are problems with inequality and welfare across countries and time?
- How are optimal taxation and public expenditure achieved?
- Why are these difficult to achieve in developing countries?

## Week 6-8: Fiscal capacity in developing countries

- What explains different tax collection patterns in developed and developing countries?
- How does large informal economy affect taxation?
- How are developing countries adopting their tax structure to increase fiscal capacities?

## Course topics (Refer to syllabus for full reading list)

### Week 9-11: Anti-poverty programs and public provision

- How should anti-poverty programs be designed?
- Form of anti-poverty programs: Cash v. in-kind? Conditional or not?
- How should public expenditures be targeted?

### Week 12-14: Organizational capacity of the state

- How to optimally allocate authority across bureaucrats and managers?
- How can public sector recruit capable and motivated individuals?
- How do public sector incentivize and monitor its workers?

## Tasks and grading scale

Two referee reports (20% of the grades)

- Aim: Critically engage with scholarly work (Guides: [here](#) and [here](#))
- 200-word summary + 1500-word critique of not-yet published academic papers
- Assess strength and weaknesses, while separating critical matters from minor issues
- For papers to referee, refer to the syllabus

Three Problem sets (20% of the grades) - Only the top two scores counts towards grade

- Solve questions testing theoretical problem-solving skills and empirical analysis skills
- Theory: Solve mathematical model and identify intuitions in plain english
- Empirics: Replicate and extend parts of an empirical scholarly paper

# Tasks and grading scale

## Research Proposal and Virtual Paper (30% of the grades)

- Develop a viable plan for a paper and build-up towards a full paper
  - 2-page research proposal outlining research questions, expected contribution, background, potential data sources and empirical strategy (10% of grade)
  - Virtual paper: Extend the proposal towards a paper with original analysis, building up to a full paper (20% of grade)
- ⇒ **Unfinished but original and exciting project >> complete, yet uninteresting project**

## Final exam (30% of the grades)

- Cumulative exam at the end of the semester
- Tests understanding of theoretical frameworks and connect it to empirical findings
- Exact mode (take-home vs in-person) TBD

## More on the virtual paper

Format of the virtual paper: Mimics many scholarly articles

- ① Introduction: State your research question and why it matters (academically / policy)
- ② Literature review: State the gap in the relevant literature, and how you fill that gap
- ③ Background: State the relevant information on the context that you are studying
- ④ Data: Provide explanation on data source and why it works in your favor
- ⑤ Empirical strategy: Explain research methods, whether it allows for causal inference, and potential limitations in that method
- ⑥ Preliminary results: Highlight main findings (causational relationships)
- ⑦ Mechanism and falsification tests: State what explains your main results, and what plans you have to address potential factors that may nullify your findings

## Important deadlines

- March 31st: Referee report 1 due
- April 14th: Problem set 1 due
- April 28th: Research proposal due
- May 12th: Problem set 2 due
- May 19th: Refere report 2 due
- June 2nd: Virtual paper due
- June 9th: Problem set 3 due
- Week of June 9th: Final exam (mode to be declared later)

## Changes in schedule around March

No class on March 24th and March 31st (Conferences in Europe and Thailand)

- Two make up classes will be scheduled
- One on March 30th, likely evening
- One on the week of April 6th (perhaps friday)
- Open to other schedules

## Some ground rules

Active discussions and questions!

Usage of AI: Strictly for assistive purposes

- Allowed for proofreading, coding support
- Do not generate complete assignments using AI
- All writings, especially virtual paper, must be original
- To prove originality, regular discussion of your ideas will suffice

Rules regarding collaboration

- You are not allowed to collaborate on referee report!
- Others: Collaboration is allowed, but under following conditions
- Each student must write up their own assignments
- Students must clearly identify their collaborators

What is Public Economics?

# Public economics study role of the state in our economic lives

State collects taxes from its citizens, and provides many goods and services

- Income tax, sales tax, social security contributions, excise tax...
- Education services, infrastructure, public safety, national defense, social safety net...
- Significant number of population works at the public sector

*Positive analysis:* Explanation of economic phenomena following government actions

- How do taxes affect (or distort) labor supply, savings, or other economic behaviors?
- How are the effects distributed across different types of individuals?

*Normative analysis:* Making judgements on how the government should act

- How progressive should the tax system be?
- To whom should we provide more public services?

In this course, we focus on three pillars of public economics: Taxation, public goods, and personnel

# Why do taxes matter

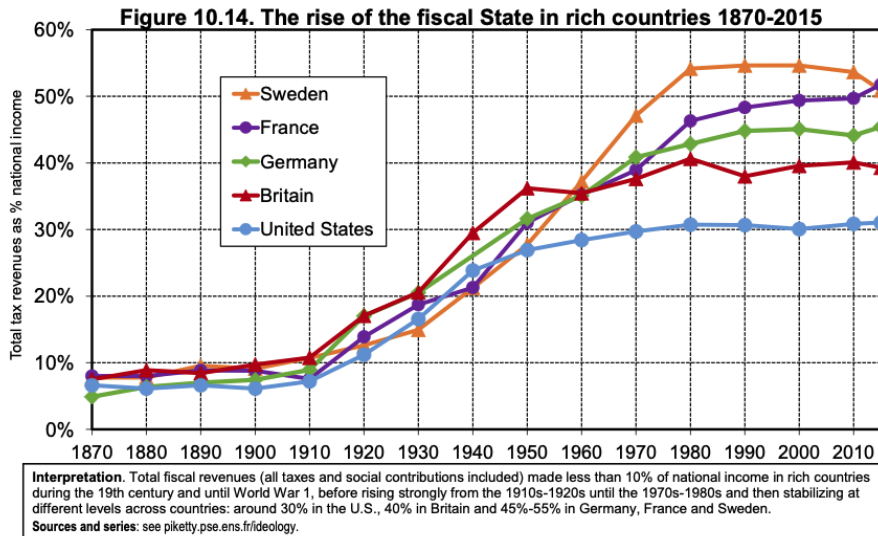
Sources of revenue for governments to finance their key services

- Finance essential public goods and services (defense, infrastructure, education, health)
- Reduce inequality through redistribution
- May come at a cost: Distort labor incentives → Fall in efficiency

Development of tax revenues and unresolved questions

- Tax share of GDP: 10% in 1910 → 30-50% today (in developed countries)
- What is the optimal design of taxation?
  - ▶ How progressive should taxes be?
  - ▶ Which taxes minimize distortions while raising needed revenue?
- Many developing countries lag behind:
  - ▶ What explains low tax collection capacity?
  - ▶ Do optimal tax prescriptions from developed countries apply here?

# Rise of the state in the developed countries



# Why are taxes enforced differently?

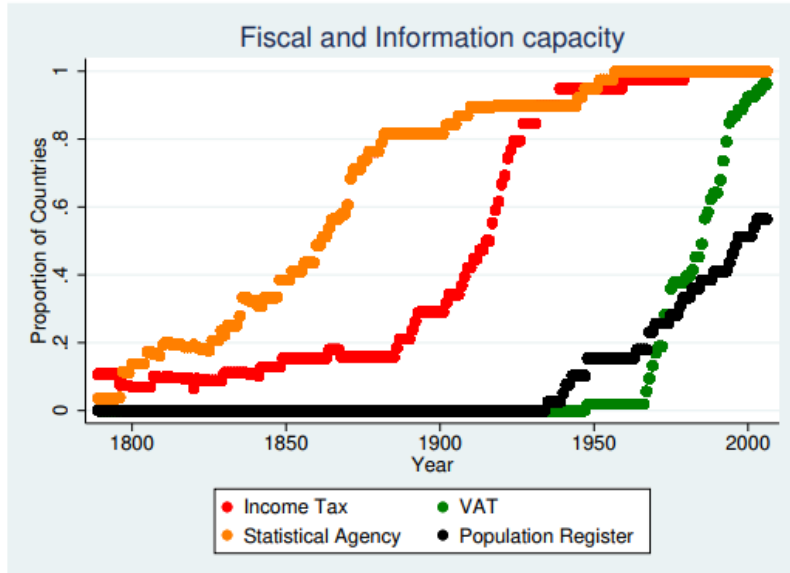
Differences in observed policy choices across place and time

- Significant progressive income taxes and sizeable public sector in developed countries
- Many developing countries rely on regressive indirect taxes (consumption, tariffs, etc.)
- Limited public services and lack of state presence in many fragile states

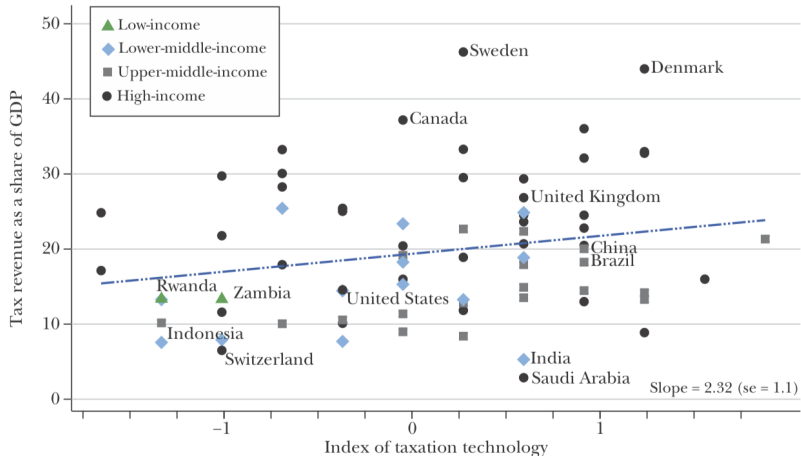
These differences originate from differences in underlying *state capacity*

- Ability of states to implement policies, collect taxes, and provide public goods
- Not all states can collect taxes digitally or have comprehensive information systems
- Not all states are fully aware of citizens' income and can target policies accordingly
- Not all states have sizeable public sectors capable of delivering policies and services

# Growth of fiscal (information) capacity in developed countries



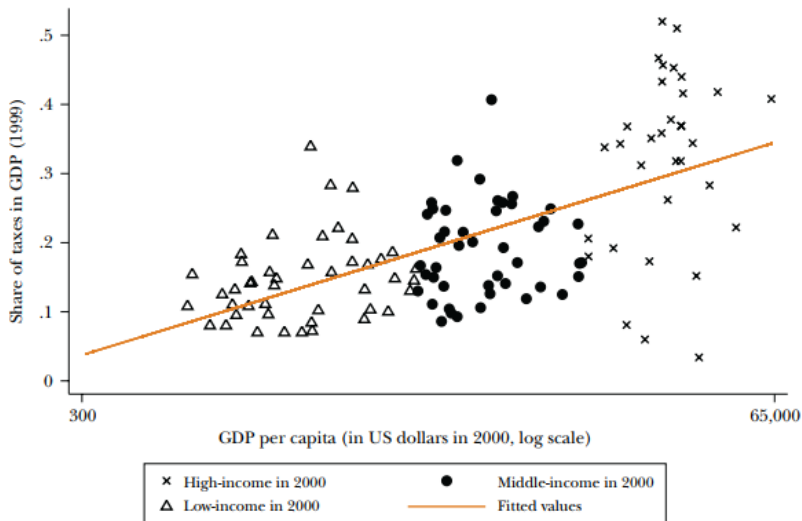
.. that is not evenly shared across countries (Okunogbe, Tourek 2024)



Index of tax tech: Use of 3rd party info, tax ID, among other technology

# Thus, drastic difference in tax intakes (Besley, Persson 2014)

Country-level Taxes and Income



# Why do public goods matter

Effective provision of public goods determines quality of life

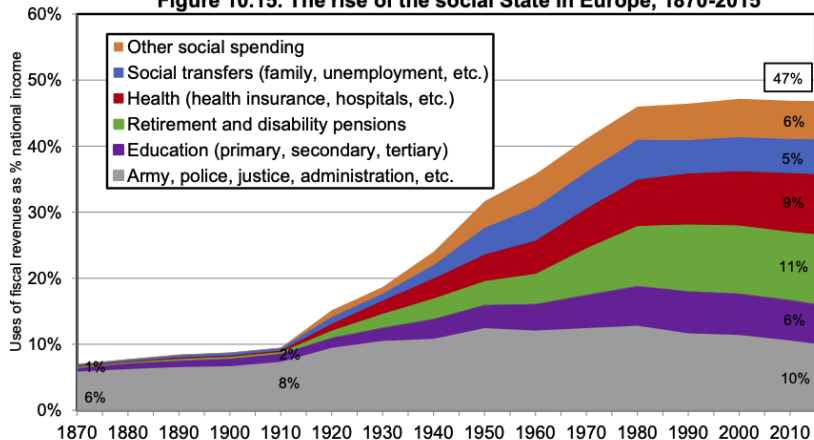
- Support markets: Law and order that enforce contracts and protect property rights
- Benefits to society beyond individual benefits: health, education, infrastructure etc.
- Underprovided by private markets: They often under-internalize social benefits

Public goods provision is central to reduction in poverty and inequality

- The rich have means to seek private alternatives, or move to 'better' locations
- The poor frequently do not
- Public spending serves to minimize discrepancy across the poor and the rich
- Huge differences across how much money is spent on such programs across countries

# Expansion of public services over time in Europe

**Figure 10.15. The rise of the social State in Europe, 1870-2015**



**Interpretation.** In 2015, fiscal revenues represented 47% of national income on average in Western Europe et were used as follows: 10% of national income for regalian expenditure (army, police, justice, general administration, basic infrastructure: roads, etc.); 6% for education; 11% for pensions; 9% for health; 5% for social transfers (other than pensions); 6% for other social spending (housing, etc.). Before 1914, regalian expenditure absorbed almost all fiscal revenues. **Note.** The evolution depicted here is the average of Germany, France, Britain and Sweden (see figure 10.14). **Sources and séries:** see [piketty.pse.ens.fr/ideology](http://piketty.pse.ens.fr/ideology).

## But significantly different even within developed countries

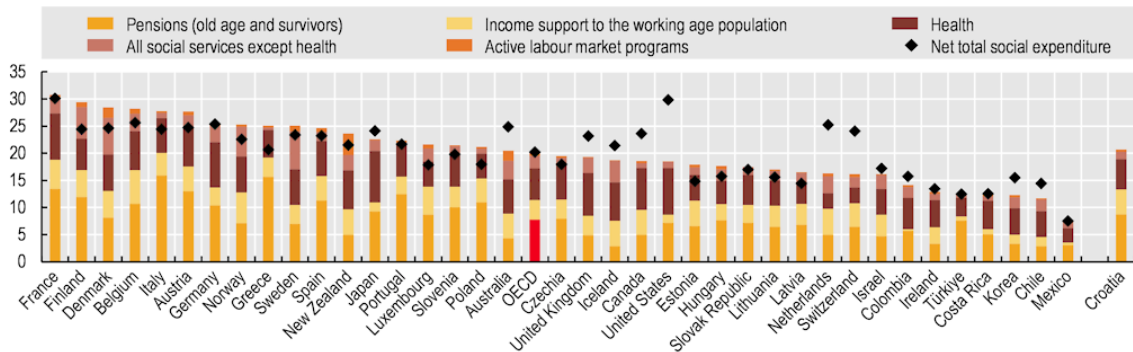
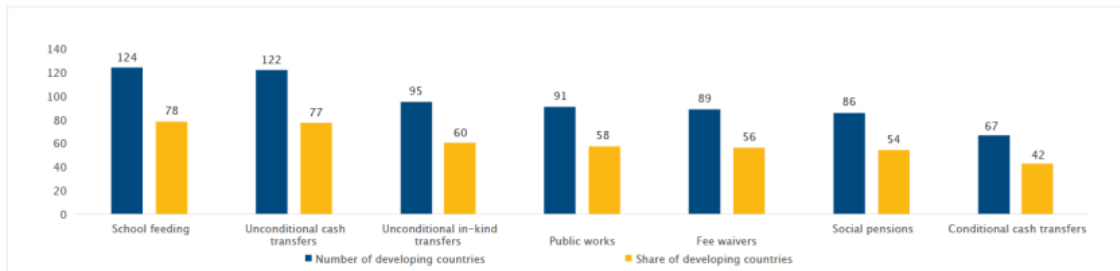


Figure: Public social expenditure as share of GDP

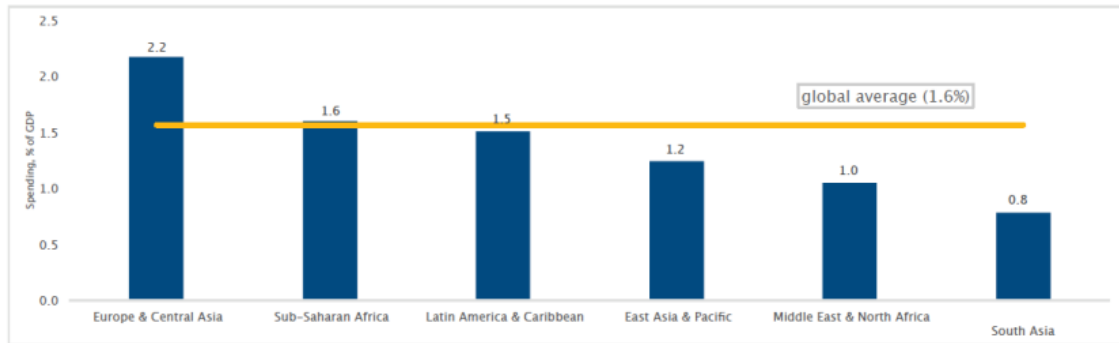
# Various public services in developing countries

**Figure 2.** Number and share of developing countries with SSN instrument



...still lag behind developed countries

**Figure 4.** Spending on SSN programs across the regions, % of GDP



# Unsettled questions in public goods / anti-poverty programs?

To whom do we target these programs? How much?

- Do you target specific populations? Or do you provide it universally?
- Do you provide support in-kind or cash?
- Do you attach conditions to income support or not?
- How do you increase take-up for welfare programs? (surprisingly low in many places)

How do you improve welfare while minimizing side-effects?

- Labor supply potentially discouraged for both recipient and the rich
  - ▶ Recipients: More earned income usually implies less support  $\rightarrow$  work effort  $\downarrow$
  - ▶ Rich: More tax collected  $\rightarrow$  less income from working  $\rightarrow$  work effort  $\downarrow$
- How do you make welfare programs politically acceptable?
- How do you design a program that also considers fiscal sustainability?

# Why public workers are important public economics

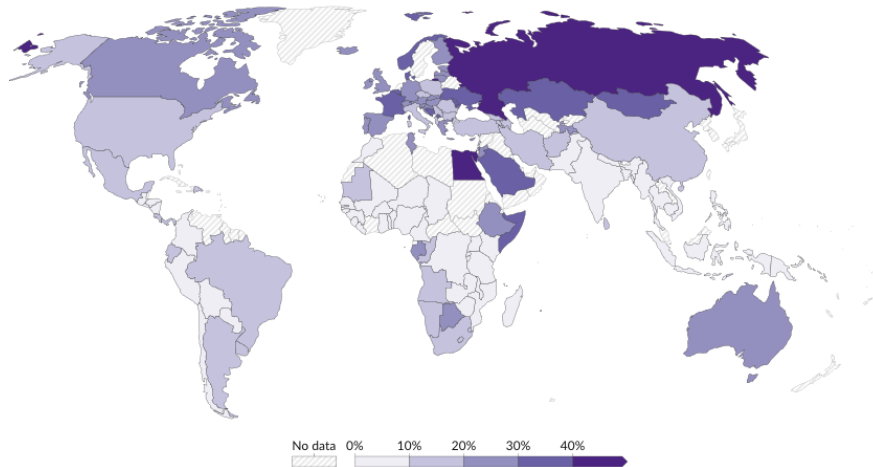
Think of them as bureaucrats working for public institutions

- They are the frontline public service providers
- They serve crucial roles for implementing various policies
- They are sizeable: 35% of all formal employment (World Bank data)

The quality of these workers matter, but also differ across contexts

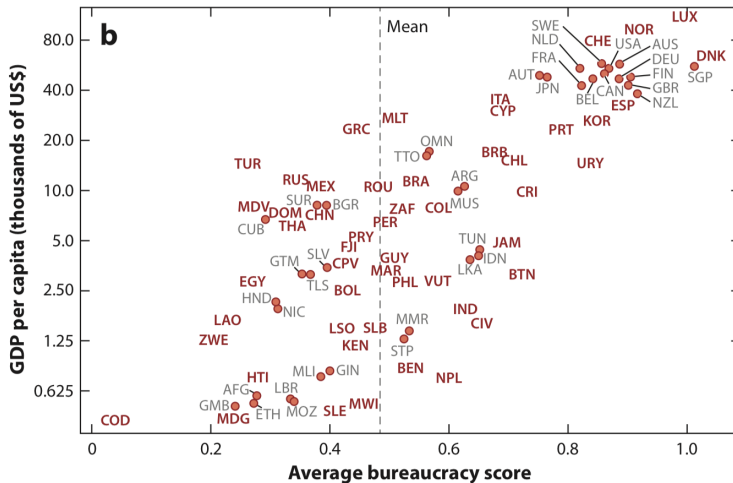
- The human capital of these workers determine the quality of service provided
- There are also large variation in the quality of public workers
  - ▶ Tend to be 'clustered' with economic development
  - ▶ Problems range from absenteeism, corruption, and lack of capabilities

# Public sector takes up huge share of employment globally



**Figure:** Share of public sector employment in total employment, 2019 (Our World in Data)

# Development and bureaucratic quality co-move (Besley et al 2022)



Bureaucracy score: Avg of bureaucratic recruitment and impartial administrative practices

# Finding and retaining motivated and dedicated workers

How do you persuade capable individuals to public sector?

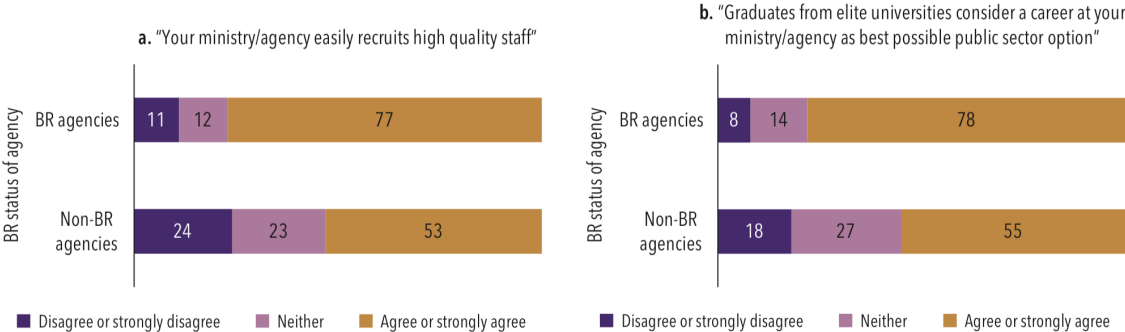
- Individuals with high abilities AND prosocial motives are ideal targets
- Public sectors in many contexts (not all!) provide lower and inflexible incentives
- Thus, smaller applicant pool and less qualified Individuals (Dal Bó et al 2013)
- Higher pay may attract individuals who apply 'just for the money' (Dessserano 2015)

How do you incentivize and monitor existing public workers?

- Public workers' interests may not always align well with the public
- Many places tried monitoring and incentive mechanisms, with mixed results
- Incentives may increase effort, but also more abuse of discretions (Balan et al 2022)
- Consistent monitoring is costly and potentially discourages efforts

# Quality of bureaucrat recruits vary across organizations

**Figure 2.21** Perceptions of the quality of recruits between BR and non-BR agencies in Indonesia

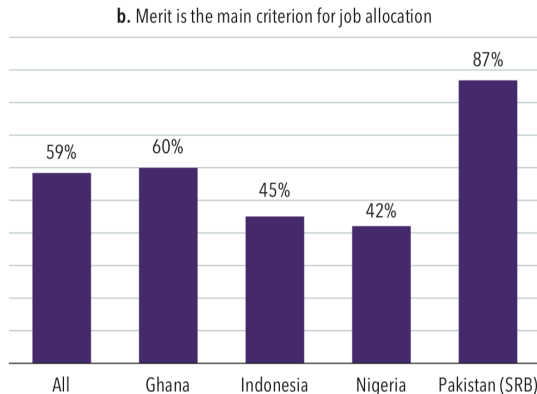
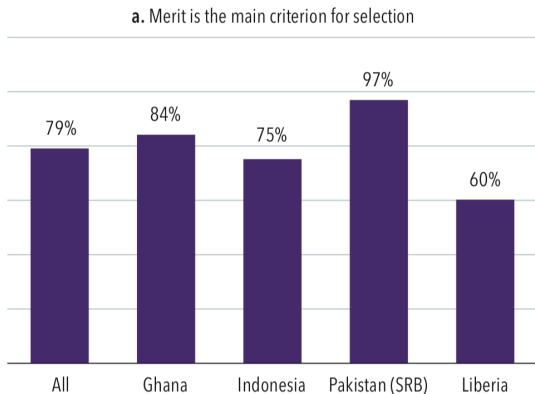


Note: BR = strategically important Bureaucracy Reform agencies.

(Source: World Bank Group (2024), *Innovating Bureaucracy for a More Capable Government*)

# Task allocation not by merit: Possible disincentives

■ ■ **Figure 3.1** Merit is generally the main criterion for selection but less so for job allocation



(Source: World Bank Group (2024), *Innovating Bureaucracy for a More Capable Government*)

# We will often use mathematical models in our discussions

Models are a simplification of reality that help us think about complex policy questions

- **Derive predictions:** What happens to people's behavior when we change tax rates?
- **Guide empirics:** What should we measure? What comparisons are informative?
- **Welfare analysis:** Does the policy make society better off/equitable?

Most models in this course share common elements:

- 1 **Agents:** Who makes decisions? (Individuals, firms, government)
- 2 **Objectives:** What do agents care about? (Utility, profits, social welfare)
- 3 **Constraints:** What limits their choices? (Budget, technology, information)
- 4 **Environment:** What are the rules? (Tax system, market structure, policies)
- 5 **Equilibrium:** What happens when everyone optimizes?

# Constrained optimization: The workhorse of economic modeling

Many economic problems can be written as constrained optimization:

$$\max_{x_1, x_2, \dots, x_n} f(x_1, x_2, \dots, x_n)$$

$$\text{subject to (s.t.): } g(x_1, x_2, \dots, x_n) \leq c$$

- **Objective function**  $f(\cdot)$ : What the agent wants to maximize
- **Choice variables**  $x_1, \dots, x_n$ : What the agent can control
- **Constraint**  $g(\cdot) \leq c$ : Limits on feasible choices

How calculus helps us solve this:

- Take derivatives:  $\frac{\partial f}{\partial x_i}$  tells us how  $f$  changes with small change in  $x_i$
- At optimum: Marginal benefit of increasing  $x_i$  = Marginal cost (from constraint)
- First-order conditions (FOCs): Set derivatives equal to zero (or use Lagrangian)
- Solve system of equations to find optimal  $x_1^*, \dots, x_n^*$

# The Lagrangian method for constrained optimization

The problem:

$$\max_{x_1, \dots, x_n} f(x_1, \dots, x_n) \quad \text{subject to: } g(x_1, \dots, x_n) = c$$

The Lagrangian approach: Convert constrained problem into unconstrained one

$$\mathcal{L}(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) + \lambda [c - g(x_1, \dots, x_n)]$$

where  $\lambda$  is the **Lagrange multiplier**

Solution method for  $x_i^*$  and  $\lambda^*$ : Take derivatives and set equal to zero

- FOC for  $x_i$ :  $\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f}{\partial x_i} - \lambda \frac{\partial g}{\partial x_i} = 0$
- FOC for  $\lambda$ :  $\frac{\partial \mathcal{L}}{\partial \lambda} = g(x_1, \dots, x_n) - c = 0$  (ensures constraint holds)
- Solve system of  $n + 1$  equations for  $x_1^*, \dots, x_n^*, \lambda^*$

Interpretation:  $\lambda^*$  = how much objective improves if constraint relaxed by one unit

## Example: Optimal tax to maximize social welfare

Objective: Maximize social welfare function  $W(\cdot)$

$$\max_t W(U_1(t), U_2(t), \dots, U_n(t))$$

where  $U_i(t)$  is utility of individual  $i$  given tax rate  $t$

Constraint: Government budget constraint (revenue needed for spending  $G$ )

$$G \leq \sum_{i=1}^n t \cdot w_i \cdot \ell_i(t)$$

Trade-off

- Higher  $t$  allows more redistribution (equity)
- But reduces labor supply  $\ell_i(t)$  and distorts economy (efficiency)
- Optimal tax balances these concerns

# From models to empirics: Why we need both

Models provide structure to empirical thought

- What parameters matter? (e.g., labor supply elasticity, social preferences)
- What would we observe if the model is correct?
- How to extrapolate from one setting to another

Data are used to test and discipline models

- Do people actually respond to taxes as predicted?
- How large are the responses? (quantitative magnitudes)
- Are there important factors the model missed?

This course emphasizes *both* theoretical frameworks and empirical evidence

## We will also use a lot of data and softwares

Working with numerical data involves working with statistical softwares

- Process raw data: Excel, GIS programs (maps to numbers), OCR (digitalize text data)
- Configure the data to a clean estimation-friendly form
- Then run estimation to obtain results and make rigorous claims
- The last two are done with Python, R, Stata ...

This involves many skill sets

- Understanding statistical inference
- Interpreting intuitions from the numerical results
- Patience (this is about 70-80%)

## Next week

We will cover classical public economics topics from week 3

- We plan to cover roughly 6 papers each course
- I will assume that you have read at least the abstract and introduction of each papers
  - ▶ Ideally, try to read from cover to cover for 1-2 papers that interest you the most

What we will do: Econometrics

- Cover applied econometrics theory and technics required to understand the papers
- We will go over regression methods: OLS, Instrumental Variables, Panel regression, Difference-in-Differences, Regression discontinuity
- Then, we will try demonstrations (demo files to be uploaded)
  - ▶ If you have a laptop with R or Stata, I highly recommend bringing it to class